A Mellin-Barnes closed form for a pass sum in Kim and Thorne (1991)

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1. The object

Fix real parameters 0 < ξ < 1 and r > 0, and let u > 0. For each integer N ≥ 1 put

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w_N = ξ^N,   a_N = r^N (1 + 4 w_N + w_N^2),   b_N = ξ^(-N) - 1 (so b_N > 0),   h(y)
= (1 + y)^(-3) for y > -1,

Ψ(u) = sum over N ≥ 1 of a_N h(u b_N).
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Write L = ln(1/ξ) > 0 and p = ln(r)/L, so r = ξ^(-p). The number p may be any real number; p < 0 exactly when r < 1. For complex z and an integer j ≥ 0, C(z, j) = z (z - 1) ... (z - j + 1)/j!.

Convergence. For u > 0 the N-th term equals (r ξ^3)^N (1 + 4 w_N + w_N^2)/(u + (1 - u) w_N)^3, which behaves like (r ξ^3)^N/u^3 as N grows. The sum therefore converges exactly when r ξ^3 < 1, that is, when p < 3.

2. Where it comes from

Kim and Thorne, Phys. Rev. D 43, 3929 (1991), eq. 67, write a renormalized stress-energy tensor as a sum over N of terms of this shape and leave the sum unevaluated. Ψ is that sum after a change of variables. The physical reading of the model belongs to its authors; nothing here rests on it or claims it.

In the paper's symbols, r = (b/2D) ξ^(-2) and u = dT/c, with dT the coordinate time left before the Cauchy horizon of their eq. 63 and c = ρ^2 D/(2 X (D - X)); a prefactor in D, X and c stands outside the sum. This is algebra on the printed eq. 67.

3. The representation

H(s) = integral from 0 to infinity of y^(s-1) h(y) dy = Γ(s) Γ(3 - s)/2 = π (1 - s)(2 - s)/(2 sin(π s)), for 0 < Re s < 3 (a Beta integral). Φ(s) = sum over N ≥ 1 of a_N b_N^(-s), absolutely convergent for Re s > p. For p < 3, u > 0 and any c with max(0, p) < c < 3:

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Ψ(u) = (1/(2 π i)) times the integral along Re s = c of H(s) Φ(s) u^(-s) ds.
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Φ continues to a meromorphic function on the whole plane, Φ(s) = sum over n ≥ 0 of Q_n(s) ρ_n(s), with ρ_n(s) = 1/(e^(L(n + s - p)) - 1) and Q_n(s) = sum over m = 0 to min(2, n) of c_m C(s

+ n - m - 1, n - m), where (c_0, c_1, c_2) = (1, 4, 1). Its poles are simple, at s_{n,k} = p - n - 2 π i k/L (n = 0, 1, 2, ...; k any integer), with residue Q_n(s_{n,k})/L (a zero residue means no pole there).

4. What "closed form" means here (claims C1 to C4)

Not a finite formula in elementary functions. It means the exact integral of section 3, valid for every u > 0, together with two residue expansions that are convergent series of explicit terms (Gamma-type factors, the polynomials Q_n, and finite sums of geometric sums).

- **C1, large u.** For u > ξ/(1 - ξ): Ψ(u) = sum over l ≥ 0 of (-1)^l C(l + 2, 2) u^(-(3 + l)) Φ(3 + l). The series converges there and diverges for 0 < u < ξ/(1 - ξ), so ξ/(1 - ξ) is its exact radius.
- **C2, small u, p not an integer.** For 0 < u < 1: Ψ(u) = sum over l ≥ 0 of (-1)^l C(l + 2, 2) Φ(-l) u^l + (1/L) sum over n ≥ 0 and all integers k of H(s_{n,k}) Q_n(s_{n,k}) u^(-s_{n,k}), both series absolutely convergent, each Φ(-l) a finite sum of geometric sums. The l-series diverges for u > 1, so 1 is its radius.
- **C3, integer p.** When p is an integer, the pole of H at s = -l and the pole of Φ at s = p - n, with n = l + p, meet. Their two residues, each growing without bound as p approaches the integer, combine into one finite term that tends to a logarithm. At the integer the pole is double and gives a term F u^l plus a term G u^l ln(1/u).
- **C4, the ripple.** The terms with k not zero come in conjugate pairs, u^(-(p - n)) times u^(+2 π i k/L) and its conjugate; each pair sums to a cosine in ln u with period ln(1/ξ)/k. The poles are spaced 2 π/ln(1/ξ) apart in Im s; the ripple's period in ln u is ln(1/ξ). This corrects my October 21 email, which gave 2 π/ln(1/ξ) as the period; section 5 reports a test of that value.

The gap. For ξ > 1/2, ξ/(1 - ξ) > 1, so for 1 ≤ u ≤ ξ/(1 - ξ) neither series converges and the integral of section 3 is the closed form there.

5. Standing, at its real size

Stated as a conjecture with numerical evidence. A written proof draft (2026-09-27) gives arguments it marks proved for: the convergence condition; the integral; the continuation and poles of Φ; C1 with its exact radius; C2 for 0 < u < 1 and p not an integer, including the divergence of the l-series for u > 1; and the statements at each meeting pole in C3. It marks as open or only outlined: the full C2-type expansion at integer p (an outline, not written out); that the whole C2 series, not only its l-series, diverges for u > 1 (checked by numbers only); and a growth bound on Φ far to the left (not needed for C2). No mathematician has read the draft, so nothing here is called a theorem.

Evidence. Four pre-registered blind runs, rules and bars frozen before the points were drawn, 12 cases a run and nine points a case. 432 of 432 points matched the direct sum: largest relative difference 1.57 x 10^-15, smallest 1.46 x 10^-56, bar 10^-8. A separate sign rule in the same runs read one case wrong. A point is one (ξ, p, u); a case is one (ξ, p); the relative difference is |series - direct sum|/|direct sum|. All 432 of those points sat on the small-u side, u between 10^-4 and 10^-2 times ξ/(1 - ξ), with ξ from 1/35 to 17/25.

A fifth blind run (October 23, 2026) drew its points in the four regions the first four never reached: large u, u near 1, whole-number p (with p near a whole number), and p from 2.90 to 2.99, with ξ up to 0.943. The bars were frozen before the points were drawn, and the seed of the draw was committed by its hash before it was revealed. An independent high-precision direct sum, the referee, was good at all 123 points (117 of them scored). The closed-form routes (C1, C2, C3 and the integral of section 3) matched it at all 238 route readings against the bar of 10^-8; the largest relative difference was 5.8 x 10^-20 for the series routes and 3.4 x 10^-14 for the integral, and those are the limits of the runs' own arithmetic: a separate check with its own 30-digit code found the closed form agreeing with the direct sum to better than 10^-24 at 27 of the points. Planted wrong formulas were caught wherever they were judged.

The period, tested. On the same points, a planted copy of the routes using the period from my October 21 email ($2\pi/\ln(1/\xi)$ in place of $\ln(1/\xi)$) was rejected at all 69 series points where it was tried, including the 54 where a one-percent error in the true period was too small for the bar to judge: the wrong period moves the ripple's poles toward the real axis, so the wrong ripple is damped less and shows everywhere. That test was not blind; it measures the bar against a planted error.

What the runs are not. Agreement at points is not a proof, and says nothing between the points or outside the ranges tested (ξ above 0.943, p above 2.99, u above 5,000 $\xi/(1-\xi)$). The ripple is the part the bar sees least: where it is under about 10^-10 of the sum, a dropped ripple pair passes the bar, and in the fifth run it was judged at only 11 of 69 series points.

Novelty. Not found so far in the sources reached (the references below). The Mellin method itself is standard (Flajolet, Gourdon and Dumas, 1995).

6. Terms a reader may have found undefined, and where each is defined

- u and r: pure parameters here (section 1); their physical meaning is Kim and Thorne's (section 2 gives the change of variables).
- p: derived, $p = \ln r/\ln(1/\xi)$ (section 1). The two sides $p < 0$ and $0 < p < 3$ behave differently as u goes to 0.
- "closed form": section 4. "Two terms meet": C3. "Ripple" and its period: C4. The gap band for $\xi > 1/2$: end of section 4.
- "blind run", "bar", "relative difference", "point", "case": section 5.

References: where this sum stands in the published literature

Each citation below was matched at its DOI registry record or its arXiv page on October 23, 2026.

1. **Kim, S.-W., and Thorne, K. S. (1991).** Do vacuum fluctuations prevent the creation of closed timelike curves? Physical Review D 43(12), 3929 to 3947. DOI 10.1103/PhysRevD.43.3929.
   The source. Eq. 67 writes the stress-energy tensor as a sum over N and leaves it unevaluated; the text says "we have computed only the leading-order part".

2. **Klinkhammer, G. (1992).** Vacuum polarization of scalar and spinor fields near closed null geodesics. Physical Review D 46(8), 3388 to 3394. DOI 10.1103/PhysRevD.46.3388.
   Takes the leading term and shows the series converges; does not sum it, and notes lower-order terms it does not compute (as read in his 1992 Caltech PhD thesis, chapter 3).

3. **Visser, M. (1994).** Hawking's chronology protection conjecture: singularity structure of the quantum stress-energy tensor. Nuclear Physics B 416, 895 to 906. arXiv:hep-th/9303023.
   Takes the leading term (its eq. 34); the sum over geodesics is left in place, with no closed form.

4. **Flajolet, P., Gourdon, X., and Dumas, P. (1995).** Mellin transforms and asymptotics: harmonic sums. Theoretical Computer Science 144(1-2), 3 to 58. DOI 10.1016/0304-3975(95)00002-E.
   The standard method for sums of this shape; it does not treat this sum.

5. **Paris, R. B., and Kaminski, D. (2001).** Asymptotics and Mellin-Barnes Integrals. Cambridge University Press. DOI 10.1017/CBO9780511546662.
   A book on the method.

6. **NIST Digital Library of Mathematical Functions, section 2.5, Mellin Transform Methods.** https://dlmf.nist.gov/2.5. Open access.

Drafted and fact-checked with an AI assistant from my own records, measurements and decisions. I read and edited every sentence and I am responsible for its contents.